

PROJECT TWO: DML and DDL Syntax Practice - Building A Relational Database

PROJECT OVERVIEW

This project serves as a comprehensive technical demonstration of Data Definition Language (DDL) and Data Manipulation Language (DML) within a Google BigQuery environment. I architected a relational database from scratch, simulating an e-commerce ecosystem with seven interconnected tables to practice schema enforcement, data integrity, and complex table relationships.

TECHNICAL SKILLS DEMONSTRATED

- Schema Design: Creating a normalized relational structure with Primary and Foreign Key constraints.
- Data Definition (DDL): Utilizing CREATE, ALTER, and DROP to manage table structures and data types.
- Data Manipulation (DML): Executing INSERT, UPDATE, and DELETE operations to maintain a dynamic dataset.
- Database Constraints: Implementing NOT ENFORCED keys (standard for BigQuery) to document logical relationships between entities.

DATABASE SCHEMA

The database consists of the following seven entities, designed to simulate a real-world retail environment:

Table	Description	Primary Key
Users	Customer profiles, contact info, and status.	user_id
Categories	Product classification and descriptions.	category_id
Products	Inventory details including pricing and brand.	product_id
Orders	High-level transaction records and totals.	order_id
OrderItems	Line-item details for every order.	order_item_id
Payments	Transactional status, card types, and IDs.	payment_id
Reviews	Customer ratings and verified purchase status.	review_id
Addresses	Shipping and primary residence management.	address_id

KEY PROJECT HIGHLIGHTS

Structural Evolution: Rather than static creation, I practiced evolving the schema as business needs change

- Dynamic Column Addition: Used ALTER TABLE to add quantity to the OrderItems table and payment_method to the Payments table after initial creation.
- Data Integrity: Implemented PRIMARY KEY and FOREIGN KEY references to map the logical flow from Users → Orders → OrderItems.
- Precise Data Maintenance: I performed targeted updates to simulate real-world data maintenance tasks:
- Customer Updates: Updated user contact information (e.g., email corrections) using specific WHERE clause filtering.
- Price Management: Modified retail pricing for specific inventory items (UPDATE) to reflect market changes.
- Record Sanitization: Utilized DELETE to remove erroneous transaction entries while maintaining referential integrity.

PROJECT CONCLUSION

This practice project established the technical rigor required to manage large-scale datasets. By manually constructing these relationships and manipulating the data within them, I developed a deep understanding of how backend data structures support frontend business analytics and reporting.